

Supplementary information to:

Proteins with multiple G protein-coupled receptor domains

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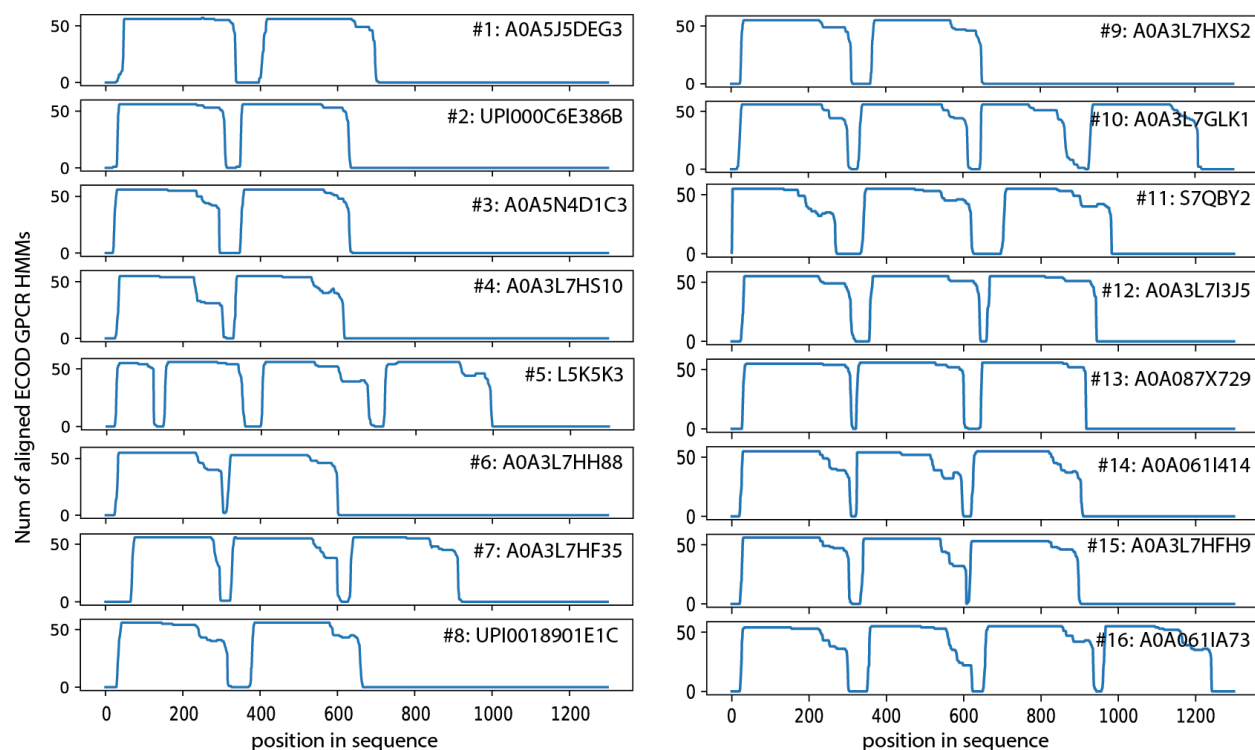


Figure 1S: The number of ECOD GPCR HMMs aligned to each position in sixteen representatives of clusters with multi-GPCRs. In each one, we can see stretches that are longer than 200 residues with multiple matches. For each panel, the number of cluster and the representative protein is listed. In each one we see multiple stretches that are candidates for GPCR segments. Notice that in some of these proteins there are long stretches that are shorter than a full GPCR (e.g., in cluster #5), as described in the text.

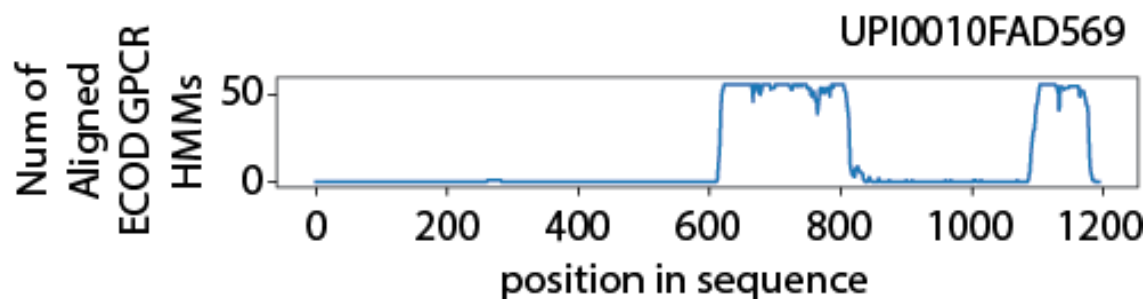


Figure 2S: The sequence UPI0010FAD569 was filtered out because there was only a single sufficiently-long segment along the sequence that matches to the ECOD GPCR HMMs (between residues 610-837). The stretch between residues 1086-1184 is not long enough. Thus, we removed this sequence, and 33 additional (shorter) sequences that were in its CD-HIT cluster.

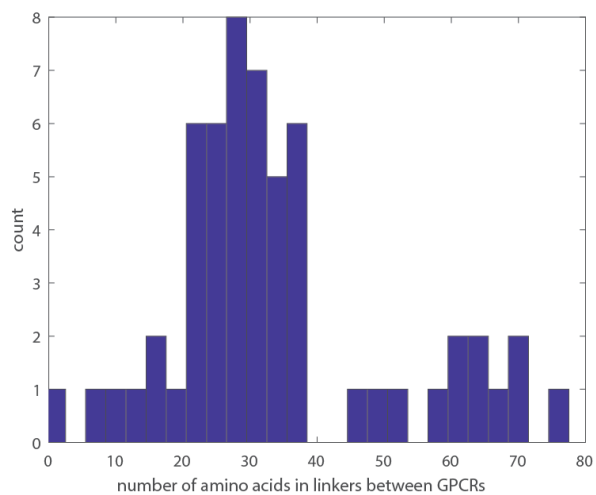
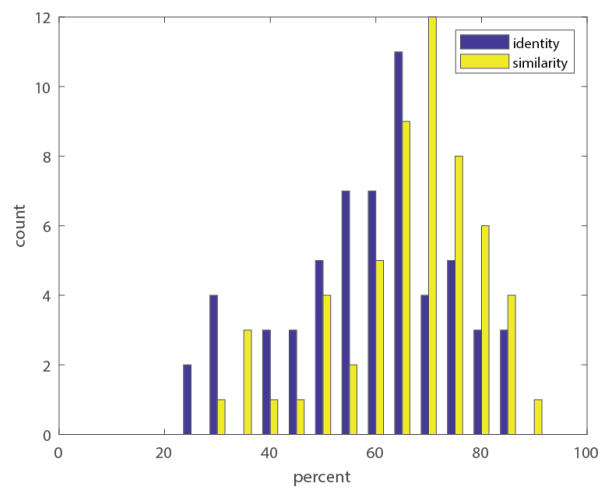
**Table
S1.**

Single-domain GPCRs, homologous to some of the multi-GPCRs reported here, which are

known to form dimers (see [3] and references therein). The data refers to specific GPCR types that are homologous to some of the multi-GPCRs discovered in this study, and which are found in the ECOD database. The ECOD IDs that correspond to each GPCR type are specified.

GPCR type	ECOD IDs
Rhodopsin	e4zwjB1 e1jfpA1 e1u19A1
Beta-1 adrenergic receptor	e4bvnA1 e2vt4A1
Beta-2 adrenergic receptor	e2r4rA2 e2rh1A4 e3kj6A2 e3p0gA1 e3sn6R4 e4ldIA3
Muscarinic acetylcholine receptor	e4dajA3 e4u15B1
Histamine H1 receptor	e3rzeA3
A _{2A} adenosine receptor	e2ydvA1 e3emIA4 e3reyA1 e3vgaA2 e4eiyA3
Angiotensin II receptor (AT1)	e4yayA1
Lysophosphatidic acid receptor	e4z34A2 e4z36A1

Figure 3S: (A) Distribution of sequence identify/similarity among the different GPCR domains within the same chain.
(B) Distribution of the number of residues in the linker regions between the different GPCR domains in the same chain.



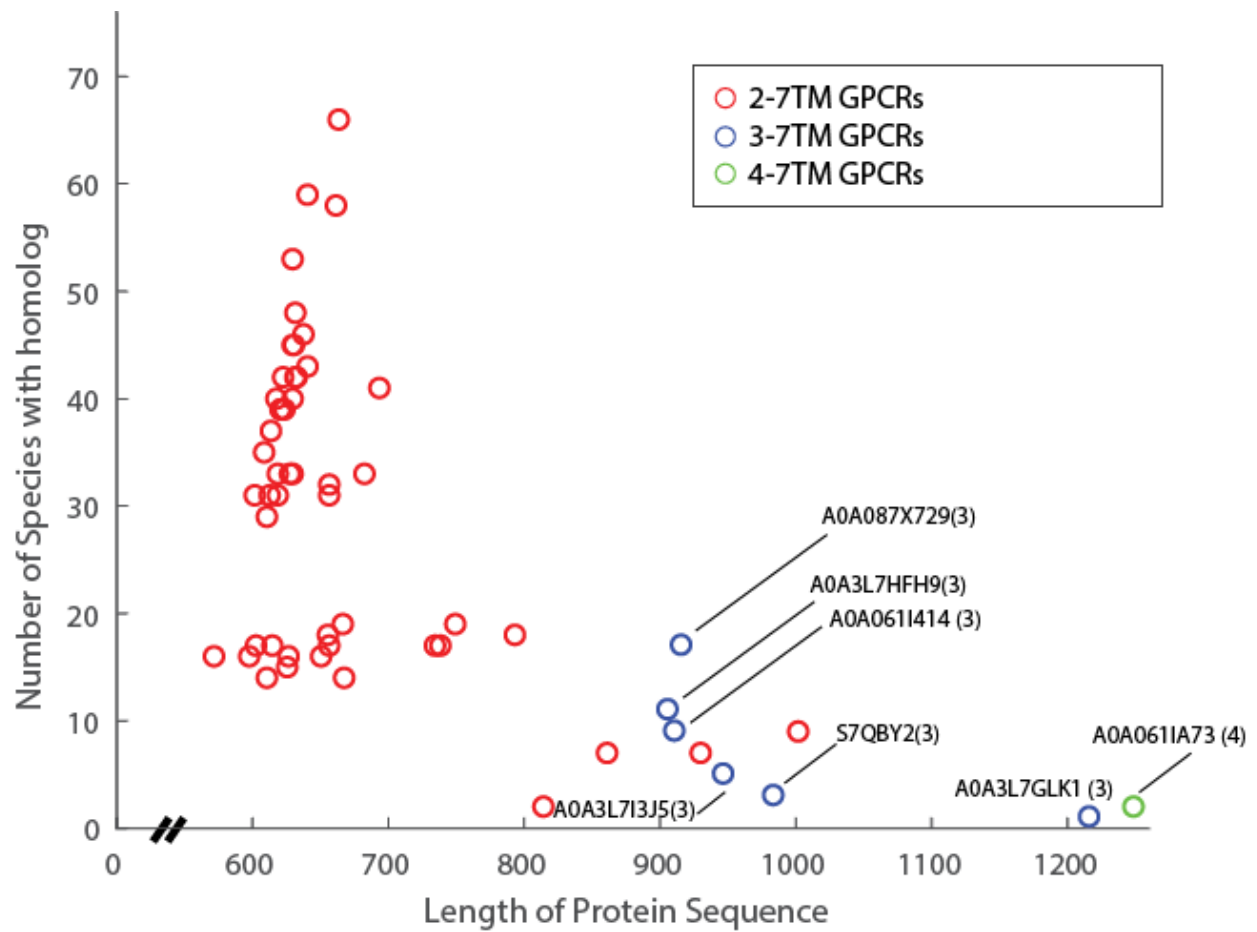


Figure 4S: The number of species with a detectable multi-PCR receptor decreases with the length of protein sequence . The number of residues in the set of 57 proteins with multi GPCR domains ranges from 571 to 1250 (x-axis), and the number of species ranges from 1 to 75 (y-axis). The identities of the 6 proteins with 3 GPCR domains, and the single protein with 4 GPCR domains are marked.

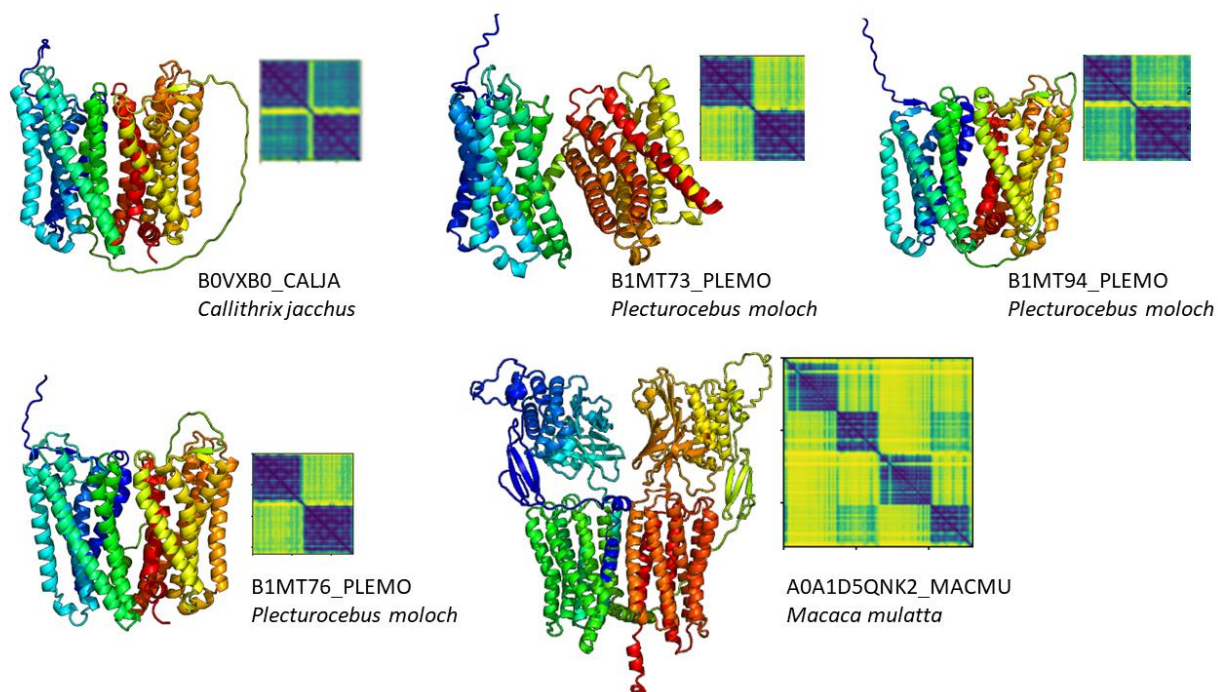


Figure 5S: The best AlphaFold models and the predicted error for the five multi-GPCR primate proteins found

Table S5. Possible agonists for some of the multi-GPCRs, inferred from literature survey of their single GPCR homologues. The multi-GPCRs shown here (column 1) are all olfactory receptors (ORs). For each OR type (column 2), the table specifies the name of the agonist reported for a single-domain GPCR (column 3), as well as its PubChem (column 4) and CAS (column 5) numbers. When available, the log EC50 value (column 6) is shown as well.

1	2	3	4	5	6	7
Multi-GPCRs	OR	Agonist	PubChem	CAS no.	log EC50	Database/Ref
A0A5N4D1C3 A0A402FIC8 L9KKD4 A0A3L7GLK1	OR4D1	androstenone	6852393	18339-16-7	n.a.	primaryodors.org
L5MBD6	OR4D2	eugenol acetate	7136	93-28-7	-5.0	[1]
A0A091CTM0 A0A485NDU3	OR8K1	nonanal	31289	124-19-6	-5.5	primaryodors.org
L9KTW2	OR8K5	(R)-carvone	439570	6485-40-1	-5.3	[2]
A0A3L7HFZ0 A0A5E4CY53	OR4K3	eugenol	3314	97-53-0	-3.3	primaryodors.org
A0A3L7HXS2	OR52D1	methyl octanoate	8091	111-11-5	-1.1	primaryodors.org

References:

1. Mainland, J.D., et al., *Human olfactory receptor responses to odorants*. Sci Data, 2015. **2**: p. 150002.
2. Cong, X., et al., *Functions of olfactory receptors are decoded from their sequence*. bioRxiv, 2020: p. 2020.01.06.895540.
3. Felce, J.H., et al., *Receptor Quaternary Organization Explains G Protein-Coupled Receptor Family Structure*. Cell Rep, 2017. **20**(11): p. 2654-2665.